



TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

FIR temperature sensor

Melexis has expanded its far-infrared (FIR) temperature sensor line with the MLX90637, the first surface-mount



device (SMD) designed for critical EV components. It enables fast, accurate, non-contact temperature monitoring for inverters, motors, and HVAC systems. Compared to traditional NTC thermistors, the FIR sensor offers faster response, higher precision, and simpler assembly. Its 50° field of view and 0.02°C resolution ensure safe operation in demanding applications like inverter busbar monitoring. The SMD format also reduces production costs versus NTC and TO-CAN solutions.

Melexis

<https://www.melexis.com>

Super temperature-compensated oscillator

Troops and systems navigating without GNSS signals face risks like navigation errors, collisions, and system disruptions, which can occur in minutes or



hours. Outages may stem from terrain, physical obstructions, or jamming and spoofing. During these periods, positioning, navigation, and timing (PNT) systems must rely on local

sensors until GNSS signals return. SiTime's Endura Super TCXO, ENDR-TTT, ensures GNSS receivers remain operational even when satellite signals are lost. It provides timing, low power consumption, and resistance to jamming and spoofing. ENDR-TTT extends holdover, maintaining timing and positioning during outages, with up to 20x greater stability and 20x greater navigational precision than quartz.

SiTime

<https://www.sitime.com>

Mini SSD card

As devices become thinner and more compact, engineers face constraints on storage capacity, reliability, and performance. The Biwin CL100 Mini SSD



addresses this by integrating the controller and flash memory into a compact LGA package, about the size of a microSD card, enabling easy expansion in ultra-thin laptops, mini PCs, and handheld gaming devices. It supports workloads such as AI processing, 4K video editing, and gaming while maintaining stable operation. Built with IP68 dust- and water-resistance, drop resistance up to 3 metres, and tolerance to vibration and humidity, the CL100 delivers consistent performance and data protection in mobile, creative, gaming, and data-intensive use cases.

Biwin

<https://www.biwintech.com>

Low-voltage Bluetooth LE SoC

Demand for connected healthcare wearables is rising due to remote

patient monitoring and home healthcare, pushing device makers to reduce size, extend battery life, and ensure connectivity in body-worn products. Nordic Semiconductor's nRF54LV10A is a



low-voltage Bluetooth LE SoC for small medical wearables. It can run from a single silver oxide coin cell, making it suitable for wearable biosensors, continuous glucose monitors, and similar devices that need extended battery life. It also combines low-voltage operation with Bluetooth channel sounding, enabling distance measurement, indoor positioning, and presence detection for healthcare use cases where patient location and proximity matter.

Nordic Semiconductor

<https://www.nordicsemi.com>

Battery management chipset

NXP Semiconductors announced a battery management chipset featuring hardware-based electrochemical impedance spectroscopy (EIS) and nanosecond-level synchronisation, designed to improve safety, charging speed, battery



life, and system insight in electric vehicles and energy storage systems. The solution integrates EIS into three BMS components: the cell sensing device,